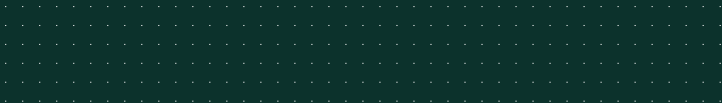


JULY 15 2025

Windows Pipe Proxy

Felix Niederwanger



Windows Pipe Proxy

A drop-in replacement for NPTP



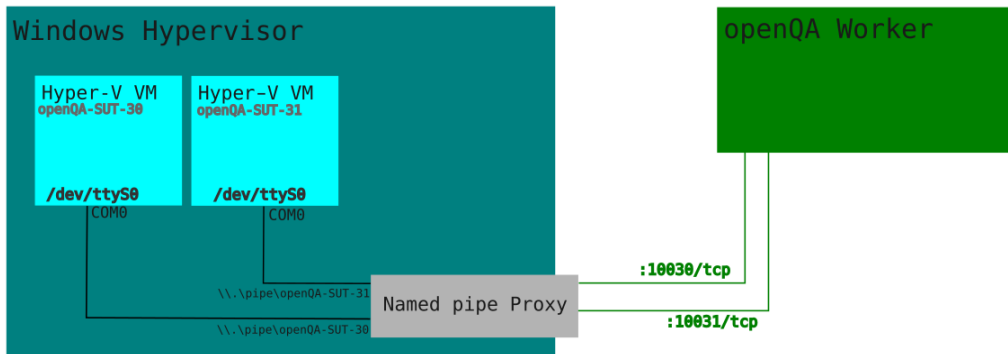
Hyper-V Serial port workflow

Problem statement

- Bridge between Serial Port of Windows VM and openQA Serial port
- Unmaintained Named Pipe TCP Proxy tool written by Shvechkov Alexey
- Last release: 2012, no official web presence anymore
- It works, but who knows for how long?
- Observed issues with more recent versions on Windows



Hyper-V Serial port workflow



A named pipe proxy connects the serial port of a Hyper-V to a openQA worker via the exposed named pipe on the Windows Hypervisor.



Windows Pipe Proxy

Suggested tool

- Drop-in replacement for NPTP
- **No openQA workflow changes required**
- TOML-based configuration file (persistent)
- Written in Rust
- Native Windows builds



Configuration

```
1 # C:\pipe-proxy.toml
2
3 ## Program configurations #####
4
5 [Plumber]
6 ReconnectAttempts = 10
7 ReconnectDelay = 1
8
9 ## Pipe configurations #####
10
11 [[Pipe]]
12 Pipe = '\\.\pipe\openQA-SUT-30'
13 Address = '0.0.0.0:10030'
14
15 [[Pipe]]
16 Pipe = '\\.\pipe\openQA-SUT-31'
17 Address = '0.0.0.0:10031'
```



Field testing

The screenshot shows a Windows 10 desktop with a blue background. In the foreground, a Hyper-V Manager window is open, displaying a list of virtual machines. Below it, a Windows PowerShell terminal window is open, showing the execution of a proxy tool.

Hyper-V Manager

File Action View Help

Hyper-V Manager

WIN-E1HRNA3UUN4

Name	State	CPU Usage	Assigned Memory	Uptime
openQA-SUT-30	Running	0%	2048 MB	00:03:15
openQA-SUT-31	Running	0%	2048 MB	00:08:19

Actions

- WIN-E1HRNA3UUN4
- New
- Import Virtual Machine...
- Hyper-V Settings...
- Virtual Switch Manager...

Checkpoints

No virtual machine checkpoints are currently displayed.

Details

No features are currently displayed.

Administrator: Windows PowerShell

```
PS C:\> .\windows-pipe-proxy.exe
Windows Pipe Proxy v0.2
\\.\pipe\openQA-SUT-30 => 0.0.0.0:10030 is listening
\\.\pipe\openQA-SUT-31 => 0.0.0.0:10031 is listening
Connected: \\.\pipe\openQA-SUT-31 <==> 10.168.192.250:39540
Connected: \\.\pipe\openQA-SUT-30 <==> 10.168.192.250:37868
```

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Field testing

Verification runs

Tool has been tested on SLES 16.1 MinimalVM using two test scenarios:

- jeos-main (Default test run)
- filesystems (Known to have issues with NPTP)



Next steps

The suggested next steps are as follows

Tooling improvements

- Create a system service for the tool
- Provide a clean installer

Deployment

- Deploy the tool to one of the Hyper-V production machines
- **Safe rollback possible because drop-in replacement**
- Test it during a full week
- Re-evaluate and user feedback



Summary

Summary

- Drop-in replacement to replace unmaintained NPTP tool
- Part of the core Hyper-V workflow
- Ready for first production field tests
- User feedback required to continue
- Current state: Build user acceptance
- Next step: Move to `os-autoinst` organisation

<https://github.com/grisu48/WindowsPipeProxy>

